AI-Powered Intelligent Home Planning & Optimization System

AI-Powered Land Optimization & Plan Generation

Group ID : 25-26J-201

BSc (Hons) degree in Information Technology

Specializing in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

Sri Lanka

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Dissertation submitted in partial fulfillment of the requirements for the Bachelor of Science (Hons) in Information Technology
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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
Sunkalpani S M D R	IT22914996	

The supervisor/s should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the supervisor: Date

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Signature of the co-supervisor: Date

ABSTRACT

Residential house planning plays a critical role in ensuring efficient space utilization, comfort, and functionality within a limited land area. However, designing a suitable house plan based on land dimensions and user requirements often requires professional expertise and manual effort. This creates a need for an automated system that can assist users in generating practical and optimized house layouts without requiring advanced architectural knowledge.

This research proposes a web-based land optimization and plan generation system that utilizes machine learning techniques and rule-based logic to generate optimized single-story residential floor plans. The system accepts user inputs such as land shape, width, depth, total area, room requirements, and front orientation. Based on these inputs, it generates a land plot diagram and an optimized floor plan that ensures efficient use of available space. The optimization process focuses on minimizing unused areas, avoiding room overlaps, improving compactness, and maintaining proper spacing for ventilation, lighting, and accessibility.

The system incorporates machine learning models such as Random Forest, Gradient Boosting, Ridge Regression, Logistic Regression, and XGBoost to evaluate layout quality and support decision-making in plan generation. These models analyze dataset features including land dimensions, buildable area, room distribution, utilization score, compactness, lighting score, and room area ratios to predict the effectiveness of generated layouts. Among the evaluated models, XGBoost demonstrated the highest performance with approximately 99% prediction accuracy.

The proposed system bridges the gap between traditional house planning methods and modern intelligent systems by providing an automated, user-friendly solution for generating optimized floor plans. It is particularly useful for individuals, architects, and developers who require quick and efficient layout suggestions during the early stages of construction planning. The system contributes to improving land utilization and simplifying the house planning process through the integration of machine learning and optimization techniques.

Keywords: Land optimization, floor plan generation, machine learning, space utilization, residential planning, XGBoost, web-based system

ACKNOWLEDGEMENT

Several individuals contributed significantly to the successful completion of this research. First, I would like to express my sincere gratitude to my supervisor, Prof. Samantha Rajapaksha and co-supervisor Ms. Adya Dissanayake for their continuous guidance, support, and valuable feedback throughout this project. I would also like to thank the Sri Lanka Institute of Information Technology for providing the necessary academic support and resources to carry out this research. The encouragement and insights provided by the academic staff were instrumental in completing this study. Furthermore, I extend my gratitude to all individuals who provided technical support and shared knowledge related to artificial intelligence, computer vision, and system development. Finally, I would like to thank my family and friends for their encouragement, patience, and support throughout this research journey.

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LIST OF ABBREVIATIONS

Abbreviation	**Full Form**
AI	Artificial Intelligence
AR	Augmented Reality
BHK	Bedroom, Hall, Kitchen
CAD	Computer-Aided Design
CNN	Convolutional Neural Network
IEEE	Institute of Electrical and Electronics Engineers
JS	JavaScript
MAE	Mean Absolute Error
PDF	Portable Document Format
RMSE	Root Mean Square Error
R^2	Coefficient of Determination
UI	User Interface
XGBoost	Extreme Gradient Boosting
YOLO	You Only Look Once
2D	Two-Dimensional
3D	Three-Dimensional

1. INTRODUCTION

Residential house planning is a critical process in the early stages of building design because the arrangement of spaces inside a house directly affects comfort, accessibility, natural lighting, ventilation, circulation, and overall land utilization. A house plan is not only a drawing that divides rooms within a boundary; it is a structured decision-making process that determines how effectively the available land can be transformed into a functional living environment. In traditional residential planning, users usually consult architects, draftsmen, or professional house designers to prepare a suitable plan according to land dimensions, land shape, room requirements, road access, and front orientation. Although professional planning provides reliable results, it can be time-consuming, costly, and difficult for general users who only need an initial concept before moving into detailed architectural design. Smart housing and artificial intelligence-based design approaches have been introduced to reduce this gap by supporting automated decision-making in residential planning [1], [13].

The proposed research component, Land Optimization and Plan Generation, focuses on generating optimized single-story residential house plans based on user-provided land details. The system accepts input parameters such as land shape, land width, land depth, total area, number of bedrooms, toilet requirement, kitchen, living room, dining area, optional carport, and front orientation. Based on these inputs, the system generates a land plot diagram and an optimized floor plan. The main purpose of land optimization is to use the available land efficiently while maintaining practical spaces around the house for ventilation, lighting, accessibility, and comfortable movement. Efficient land utilization is an important aspect of modern residential planning because poor space allocation can result in unused areas, poor indoor comfort, and reduced functionality [5].

In many small residential projects, users may own a limited land area and may not have sufficient architectural knowledge to decide how the rooms should be arranged. For example, a two-bedroom or three-bedroom single-story house requires balanced space allocation among bedrooms, bathroom, kitchen, dining area, living room, and circulation spaces. If these areas are not arranged properly, the final layout may become uncomfortable, compact in the wrong places, poorly ventilated, or inefficient in terms of land usage. Research related to building layout optimization has shown that space utilization and energy efficiency are strongly influenced by the way rooms and functional areas are organized within the available boundary [14]. Therefore, an intelligent system that can automatically analyze user requirements and generate an optimized layout can provide valuable support during the early planning stage.

This component is developed as a web-based system, not as a mobile application. The frontend is implemented using React JS and Vite, while the backend is developed using Python. The system also uses machine learning and data-processing libraries such as Scikit-learn, NumPy, Pandas, Matplotlib, and Seaborn for dataset preparation, model training, visualization, and evaluation. Scikit-learn is widely used in machine learning research and practical applications because it provides efficient tools for classification, regression, model selection, preprocessing, and evaluation [6]. In this system, multiple machine learning models are considered, including Gradient Boosting, Random Forest, Ridge Regression, Logistic Regression, and XGBoost, to predict the quality of generated layouts and support the optimization process. Random Forest is suitable for handling nonlinear relationships in structured data [8], while Gradient Boosting is effective for improving prediction performance by combining multiple weak learners [9]. XGBoost is also widely recognized as a scalable and high-performing tree boosting system for structured prediction tasks [7].

The system is specifically limited to single-story houses with two to three bedrooms. This limitation makes the research scope more focused and practical because most small residential users require compact and affordable house layouts. Depending on the selected bedroom count, other areas such as bathroom, kitchen, dining area, living room, and optional carport are adjusted dynamically. The system does not attempt to replace professional architects or engineers; instead, it provides an intelligent initial layout suggestion that users can use for early-stage planning and discussion. By combining rule-based optimization with machine learning-based quality prediction, the proposed component aims to reduce manual effort, improve land utilization, and support users in generating practical residential house plans.

1.1 Background & Literature Survey

Modern residential planning has become increasingly challenging due to limited land availability, irregular land shapes, rising construction costs, and the need for better space utilization. In many urban and semi-urban areas, land plots are becoming smaller, and users are required to make maximum use of the available area. A poorly planned residential layout can lead to wasted land, inefficient circulation, poor room connectivity, poor ventilation, and weak natural lighting. These issues can reduce the comfort and practicality of the house. Space optimization in residential architecture focuses on improving the functional arrangement of spaces while reducing unused or poorly utilized areas [2]. Therefore, intelligent planning methods are becoming important in residential design, especially for users who need quick and practical design suggestions before consulting professionals.

Traditional house planning usually depends on the experience and creativity of an architect or designer. Although this method is effective, it often requires multiple revisions, site analysis, user discussions, and manual drafting. This process may not be affordable or convenient for users who only require an initial floor plan idea. To overcome this issue, automated building layout generation methods have been studied in the field of construction automation and architectural computing. Automated layout generation systems attempt to arrange rooms and building components based on given requirements, constraints, and optimization rules [4]. Such systems can reduce manual workload and provide fast design alternatives for early-stage planning.

Artificial intelligence has also become an important research area in smart housing design. AI-based housing design systems can analyze user requirements, process design constraints, and support decision-making in layout generation [1], [13]. Instead of depending only on manual drawing, AI-supported systems can evaluate different design alternatives and identify layouts that make better use of available space. In residential planning, important constraints include land dimensions, buildable area, room count, room adjacency, orientation, accessibility, natural lighting, and ventilation. These constraints must be considered together because a layout that satisfies one requirement may fail another. For example, a compact layout may improve space utilization but may reduce ventilation if open spaces are not properly considered.

Automated floor plan generation is closely related to optimization. Optimization techniques help determine how different rooms should be arranged inside the land boundary while minimizing wasted space and avoiding invalid room placements. Studies on building layout optimization highlight the importance of balancing energy efficiency, space utilization, and functional arrangement during the planning process [14]. Similarly, research on automated floor plan design using genetic algorithms shows that computational methods can generate design alternatives by searching for better spatial arrangements according to defined constraints [15]. These findings support the idea that automated design systems can improve the efficiency of early-stage residential planning.

Land utilization is another important factor in this research domain. Efficient land utilization means that the available land should be used in a way that supports the user's functional requirements while preserving sufficient spacing for comfort and environmental quality. Data-driven approaches have been used to improve land utilization by analyzing design-related features and identifying patterns that support better planning decisions [5]. In the proposed system, land utilization is evaluated using features such as buildable area, utilization score, compactness, room area ratios, overlap detection, and lighting score. These features help the system determine whether a generated plan uses the available land effectively.

Machine learning provides strong support for this type of structured decision-making problem. Machine learning models can learn relationships between input features and output quality from prepared datasets [11]. In this research, the system uses structured data such as land dimensions, room count, buildable area, utilization score, compactness, lighting score, overlap detection, and final score. These features are suitable for machine learning because they represent measurable characteristics of a generated layout. By training models on these features, the system can predict whether a generated layout is efficient or requires improvement. Machine learning applications in floor plan generation have shown that prediction models can support automated planning and improve layout evaluation [3].

Several machine learning algorithms are suitable for layout quality prediction. Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs to improve prediction performance. It is useful for handling nonlinear relationships and reducing overfitting in structured datasets [8]. Gradient Boosting is another powerful ensemble method that builds models sequentially, where each new model attempts to correct the errors of previous models [9]. XGBoost extends gradient boosting by providing a scalable and efficient tree boosting framework, which is widely used for high-performance structured data prediction tasks [7]. These models are appropriate for the proposed system because floor plan quality depends on multiple interacting factors such as land area, room count, compactness, and overlap status.

Regression-based models are also useful in this research. Ridge Regression can be used as a baseline regression model to analyze relationships between numerical features and output scores.

Logistic Regression can be used when layout quality is represented as a classification problem, such as good layout or poor layout. The use of multiple algorithms allows the system to compare model performance and select the best-performing approach. In this research component, the models are evaluated using performance metrics such as Accuracy Score, Mean Absolute Error, Root Mean Square Error, R^2 Score, and Confusion Matrix. These metrics help measure both classification and regression performance, making the evaluation process more reliable.

Recent studies also show that deep learning is used in architectural layout analysis, especially for recognizing room regions, walls, doors, and spatial elements from images [16]. However, the proposed system does not mainly focus on recognizing existing plans from images. Instead, it focuses on generating new optimized layouts from user-provided land inputs. This is an important distinction because many existing approaches concentrate on layout recognition or visualization, while this research focuses on land optimization and plan generation. Therefore, the proposed system combines rule-based spatial allocation with machine learning-based quality prediction to generate practical residential layouts.

Overall, the literature indicates that automated house planning can be improved through the integration of AI, machine learning, optimization techniques, and data-driven decision-making. Existing studies provide strong support for using computational methods in smart housing design, space optimization, building layout generation, and floor plan evaluation [1]–[5], [13]–[16]. Based on these findings, this research component develops a web-based Land Optimization and Plan Generation system that generates optimized single-story residential floor plans. The system aims to support users by providing initial layout suggestions that improve land utilization, reduce wasted space, avoid room overlaps, and maintain important residential planning considerations such as lighting, ventilation, accessibility, and compactness.

1.2 Research gap

Although various architectural design tools and planning applications are available today, most of them require manual input, professional expertise, or advanced knowledge in Computer-Aided Design (CAD) systems. Users who only possess basic information such as land dimensions, shape, and required number of rooms often find it difficult to create a practical and efficient house plan. These tools primarily focus on visualization rather than intelligent decision-making, and they do not automatically optimize space utilization or evaluate the quality of generated layouts.

Existing systems typically address only specific aspects of house design, such as floor plan drawing, interior design, or 2D visualization. Very few systems integrate land optimization with automatic floor plan generation in a unified framework. Moreover, most available solutions lack the ability to analyze key spatial factors such as unused space, compactness, ventilation efficiency, and lighting distribution. As a result, users may generate layouts that are visually acceptable but inefficient in terms of space utilization and functionality [1].

Another significant limitation is that many automated design systems are not user-friendly for general users. They often require structured input formats, technical configurations, or professional interpretation. This creates a barrier for individuals who want a simple solution to generate house plans based on basic inputs such as land shape, width, depth, total area, and room requirements.

Furthermore, current research approaches in automated floor plan generation focus mainly on image-based recognition, deep learning-based segmentation, or object detection techniques. While these methods are effective in detecting layout components, they do not provide intelligent optimization or decision-making capabilities for generating new plans from scratch [2], [3]. There is a lack of systems that combine machine learning with rule-based optimization to generate efficient, real-world usable house layouts.

Therefore, the main research gap identified in this component is the absence of a web-based system that can automatically generate optimized single-story residential floor plans using simple user inputs while ensuring efficient land utilization. This research addresses the gap by developing a system that integrates machine learning models with optimization logic to produce practical house layouts with improved space utilization, proper room arrangement, and better environmental considerations such as ventilation and lighting.

Reference	**Functionalities of the Application**	**Technology and Techniques Used**	**Platform**	**Accuracy / Limitation**
Floor planner	Design and visualize house layouts using drag-and-drop tools	2D modeling, UI-based systems	Web, Mobile	Low (no optimization or intelligence)
Room Sketcher	Create floor plans and interior layouts	CAD-based design tools	Web, Mobile	Low (manual design required)
MagicPlan	Generate floor plans using camera input	Computer vision, AR-based measurement	Mobile	Medium (limited reasoning capability)
Planner 5D	2D home design and visualization	Deep learning, UI-based modeling	Web, Mobile	Low (no optimization logic)
Research Systems (Plan Recognition) [2], [3]	Detect rooms and layout elements from images	CNN, image processing	Research environments	Medium (recognition only, no generation)
YOLO-based Systems [4], [5]	Detect objects such as doors and rooms	Deep learning, object detection	Cross-platform	High detection accuracy, but no layout optimization
Proposed System	Generate optimized floor plans based on land inputs	Machine learning + optimization techniques	Web	High (~99% prediction accuracy with optimization)

Table 1.2.1: Comparison with existing systems and identifying gaps

1.3. Research problem

The main research problem is that users face difficulty in generating a suitable house plan based on land dimensions and room requirements without professional support. Manual planning requires time, cost, design knowledge, and several revisions. If the initial plan is poorly arranged, it can cause wasted spaces, poor accessibility, weak ventilation, low lighting, and inefficient land use.

For small residential houses, especially two- to three-bedroom single-story houses, users need a quick and practical way to understand how their land can be used. However, generating a good plan requires considering many factors at the same time. These include land shape, land width, land depth, total area, room count, room sizes, front orientation, carport requirement, compactness, lighting, and overlap detection.

Therefore, the research problem can be stated as follows:

How can a web-based system generate an optimized single-story residential house plan using land dimension, room requirements, and machine learning-based quality prediction?

The challenge is not only to draw rooms on a land plot but also to produce a plan that is usable, compact, non-overlapping, and suitable for the available land area.

1.4. Research Objectives

1.4.1. Main Objective

The main objective of this research component is to develop a web-based Land Optimization and Plan Generation System that automatically generates optimized single-story house plans according to land dimensions, land shape, room requirements, and front orientation.

The system aims to improve land utilization, reduce wasted areas, maintain proper spacing around the house, and generate practical two- to three-bedroom house layouts with key areas such as bedrooms, bathroom, kitchen, dining area, living room, and optional carport.

1.4.2. Specific Objectives

1. To collect and prepare land and layout-related data using features such as land dimensions, buildable area, room count, utilization score, overlap detection, compactness, lighting score, room ratios, and final score.
2. To design a plan generation logic that produces single-story layouts based on land shape, width, depth, total area, room requirements, and front orientation.
    - The system applies rule-based spatial allocation to ensure rooms are placed efficiently without overlap and within the available land boundary.
3. To apply machine learning models such as Gradient Boosting, Random Forest, Ridge Regression, Logistic Regression, and XGBoost for layout quality prediction.
    - These models evaluate the generated layouts based on optimization features such as space utilization, compactness, and lighting efficiency.
4. To generate land plot diagrams and optimized floor plans as system outputs.
    - A weighted scoring mechanism is used to evaluate layout quality, giving higher importance to critical factors such as space utilization and overlap prevention.
5. To evaluate the system using performance metrics such as Accuracy Score, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), R^2 Score, and Confusion Matrix.
6. To develop an interactive web interface using React JS and Vite for user-friendly input and visualization of generated plans.

2. METHODOLOGY

2.1 Understanding the key pillars of the research domain

The main pillars of this research component are **machine learning, land optimization techniques, automated plan generation logic, and data preprocessing with feature engineering**. These components work together to generate efficient and practical residential floor plans based on user-provided land details.

2.1.1 Machine Learning

Machine learning is used to evaluate and predict the quality of generated floor plans. Since the system operates on structured data such as land dimensions, room count, utilization score, compactness, and lighting score, machine learning models are suitable for identifying relationships between these features and the overall layout quality.

The system uses several machine learning algorithms, including Gradient Boosting, Random Forest, Ridge Regression, Logistic Regression, and XGBoost. These models are trained using prepared datasets to predict whether a generated layout is efficient or requires further optimization.

Among these models, ensemble-based methods such as Random Forest and XGBoost provide higher accuracy because they can capture complex relationships between multiple features. The prediction results help the system improve layout decisions and ensure that the final plan meets optimization criteria.

2.1.2 Optimization Techniques

Land optimization is a key component of the system, focusing on maximizing the efficient use of available land while maintaining functional and comfortable living spaces. The optimization process ensures that all rooms are properly arranged without overlap and that unnecessary or wasted spaces are minimized.

Several optimization criteria are considered during the plan generation process:

• **Space Utilization:** Ensures that the available land area is used effectively without leaving large unused spaces

• **Compactness:** Measures how efficiently rooms are arranged together to reduce wasted space

• **Overlap Detection:** Prevents rooms from overlapping or exceeding land boundaries

• **Lighting Score:** Evaluates whether the layout supports natural light distribution

• **Accessibility:** Ensures proper movement paths between rooms

These optimization techniques are implemented using rule-based logic combined with machine learning predictions to improve the quality of generated plans.

2.1.3 Plan Generation Logic

The plan generation logic is responsible for creating the actual floor plan layout based on user inputs. The system generates single-story residential plans that include key areas such as bedrooms, bathrooms, kitchen, dining area, living room, and optional carport.

The process follows a structured approach:

- The land area is divided into usable regions based on dimensions
- Required rooms are allocated space based on predefined size rules
- Rooms are arranged sequentially to avoid overlap and maintain accessibility
- Layout adjustments are made based on optimization scores
- The final plan is validated using the machine learning model

The system dynamically adjusts room sizes and positions depending on the number of bedrooms selected (2–3 rooms). This ensures flexibility while maintaining a consistent and practical layout structure.

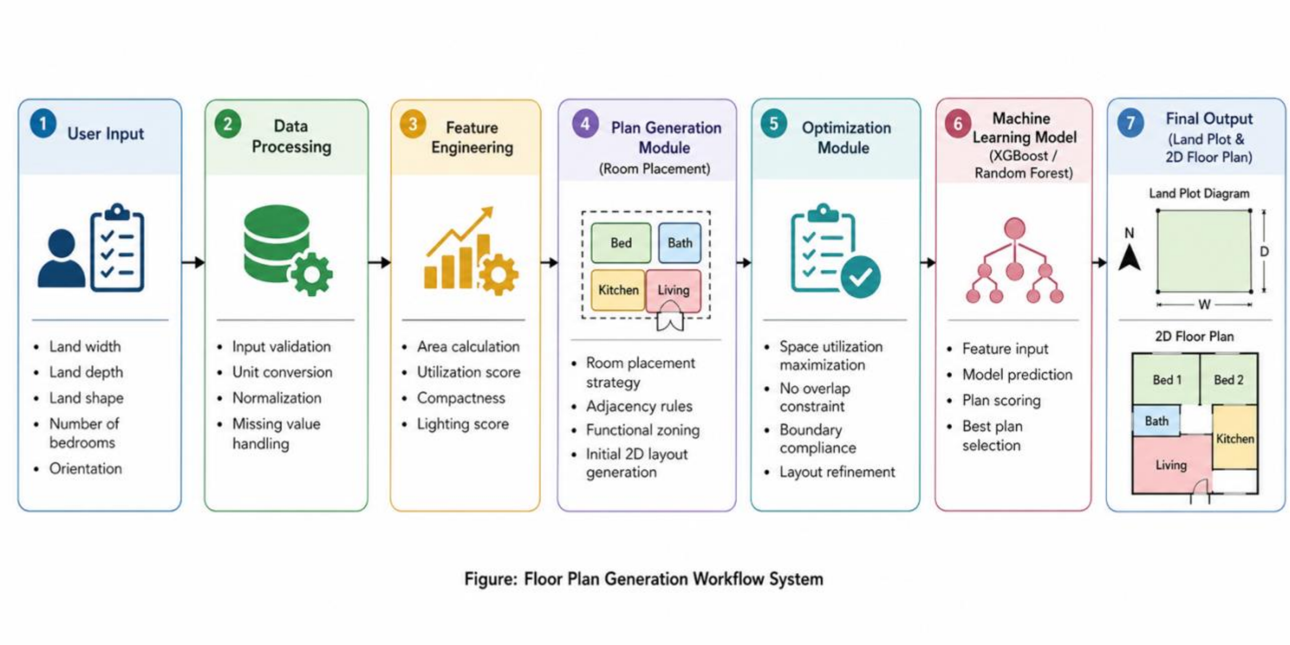

Figure 1:Floor plan generation workflow

2.1.4 Data Preprocessing and Feature Engineering

Data preprocessing is used to prepare the dataset for machine learning model training. The dataset includes features such as:

- Land dimensions (width, depth, total area)
- Buildable area
- Room count
- Utilization score
- Overlap detection
- Compactness
- Lighting score
- Room area ratios
- Final score

Feature engineering is applied to transform raw inputs into meaningful values. For example:

- Land width × depth → Total area
- Room count → Required space estimation
- Layout structure → Compactness calculation
- Window placement → Lighting score estimation

These processed features are used as input for machine learning models to predict layout quality and guide optimization decisions.

2.2 Land Optimization and Automated Plan Generation Approach

The proposed system is designed to process land-related input data, analyze the available space using machine learning and optimization techniques, and generate an optimized single-story house plan through a web-based application. The implementation process involves several stages, including user input collection, data preprocessing, feature extraction, model prediction, land optimization, and 2D floor plan generation. Each stage is carefully designed to ensure that the system uses the available land efficiently and produces a practical layout with proper room arrangement, ventilation, lighting, accessibility, and reduced wasted space.

The system accepts land shape, width, depth, total area, room requirements, and front orientation as user inputs. Based on these details, it identifies the buildable area and generates a suitable layout for two- to three-bedroom single-story houses. The generated plan includes key areas such as bedrooms, bathroom, kitchen, dining area, living room, and optional carport. Finally, the system outputs a land plot diagram and an optimized floor plan for the user.

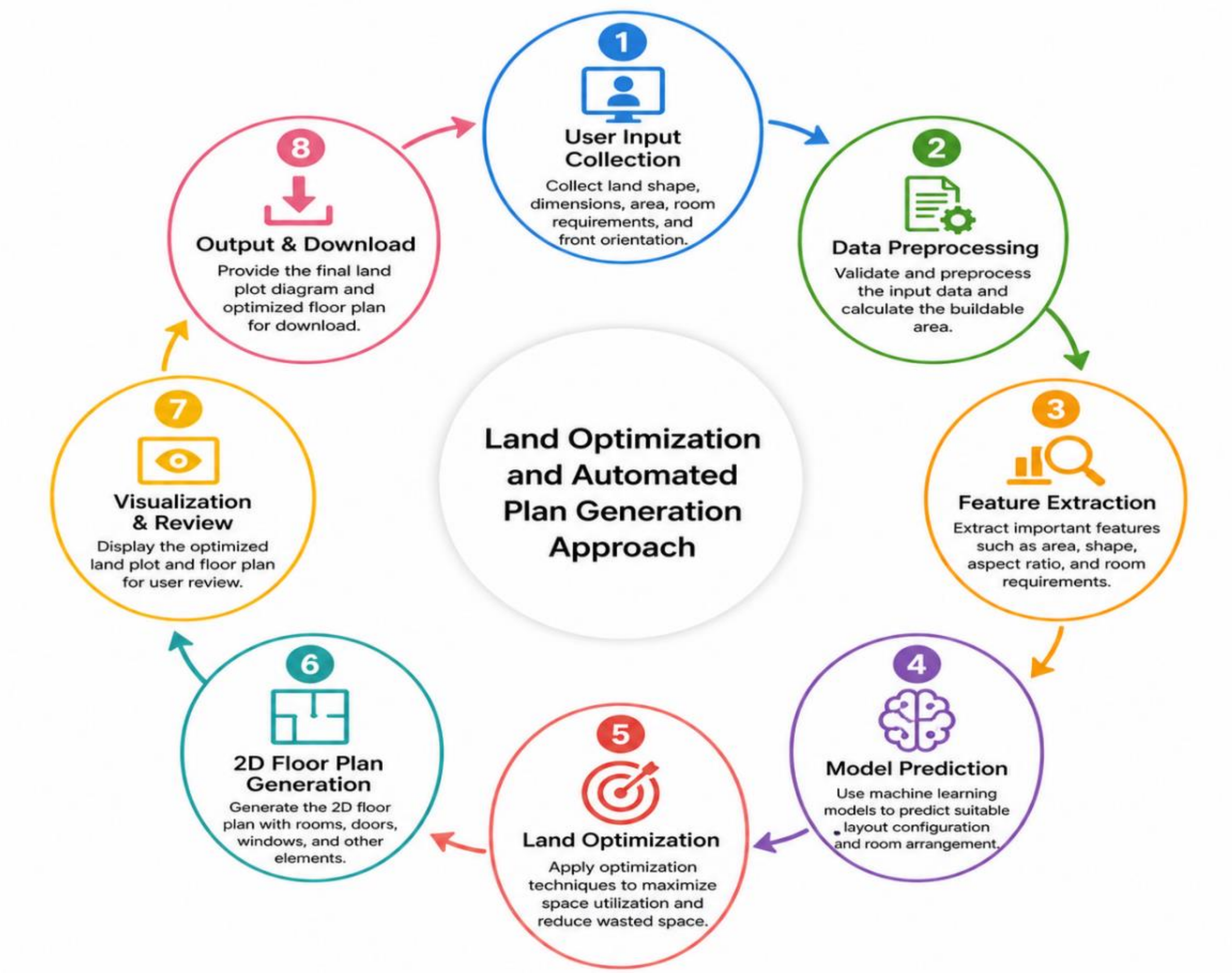

Figure 2: Methodology of the approach

2.2.1 Data Collection

The data used for this component is based on residential land and layout-related attributes. The main user inputs are land shape, width, depth, total area, bedrooms, toilets, kitchen, living room, dining room, carport, and front orientation .

The system supports three land shapes: square, rectangle, and L-shape. Width and depth are used to calculate land size and buildable area. Room requirements are used to determine the required internal layout. Front orientation is used to decide access direction and room arrangement.

2.2.2 Methodology

The implementation of the proposed land optimization and plan generation system follows a structured and systematic methodology to ensure efficiency, accuracy, and practicality in generating optimized residential layouts. The overall process is designed to transform user-provided land details into a feasible and well-organized single-story house plan while maximizing land utilization and minimizing wasted space.

The process begins with input data preprocessing, which is essential for preparing the user-provided land details for further processing. At this stage, inputs such as land shape, width, depth, total area, and room requirements are validated to ensure correctness and consistency. Any invalid or unrealistic values are filtered out or adjusted to maintain system reliability. The system then standardizes the input data into a structured format, enabling smooth processing in subsequent stages. Additionally, derived values such as buildable area are calculated based on the given dimensions, which helps in determining whether the requested layout can fit within the available land space.

Following preprocessing, the system performs feature extraction and dataset preparation. Important features such as land dimensions, room count, utilization score, overlap detection, compactness, lighting score, room area ratios, and final score are generated. These features represent the quality and feasibility of a layout and are used for both optimization and machine learning prediction. Feature engineering plays a critical role in converting raw input data into meaningful values that can guide the plan generation and evaluation process effectively.

Once the dataset is prepared, it is divided into training, validation, and testing datasets. The training dataset is used to train the machine learning models to understand relationships between land features and layout quality. The validation dataset is used during training to fine-tune model parameters and prevent overfitting. The testing dataset is used to evaluate the final model performance on unseen data, ensuring that the system can generalize well to new user inputs. This structured data splitting approach ensures that the system remains robust and reliable.

The next stage involves the application of machine learning models for layout quality prediction. Instead of relying on a single model, multiple algorithms such as Gradient Boosting, Random Forest, Ridge Regression, Logistic Regression, and XGBoost are implemented and evaluated.

These models are selected because they are well-suited for structured tabular data and can effectively capture relationships between multiple input features. During training, each model learns patterns that indicate whether a generated floor plan is efficient and feasible.

During the model training phase, hyperparameter tuning and optimization are performed to improve prediction accuracy. Parameters such as learning rate, tree depth, number of estimators, and regularization factors are adjusted to identify the best-performing configuration. Evaluation metrics such as Accuracy Score, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), R^2 Score, and Confusion Matrix are used to measure model performance. Continuous monitoring of training and validation results ensures that the selected model achieves high accuracy while avoiding overfitting. The final system achieves approximately 99% prediction accuracy, demonstrating strong performance in evaluating layout quality.

After model training, the system proceeds to the land optimization and plan generation phase. In this stage, rule-based logic and optimization techniques are applied to arrange rooms within the available land space. The system ensures that rooms do not overlap, remain within the land boundary, and are arranged in a compact and accessible manner. It also considers proper spacing for ventilation and natural lighting, improving the overall usability of the layout. Based on the number of bedrooms selected (two or three), the system dynamically adjusts the sizes and arrangement of other areas such as the kitchen, dining area, living room, and bathroom.

The generated layout is then evaluated using the trained machine learning model to determine its quality. If the layout does not meet the required standards, the system iteratively adjusts the arrangement to improve space utilization and compactness. This combination of rule-based generation and machine learning evaluation ensures that the final output is both practical and optimized.

Finally, the system produces the output visualization, which includes a land plot diagram and an optimized 2D floor plan. These outputs are generated and displayed through the web-based interface developed using React JS and Vite. The diagrams clearly illustrate room placement, dimensions, and overall layout structure, allowing users to easily understand and interpret the generated plan.

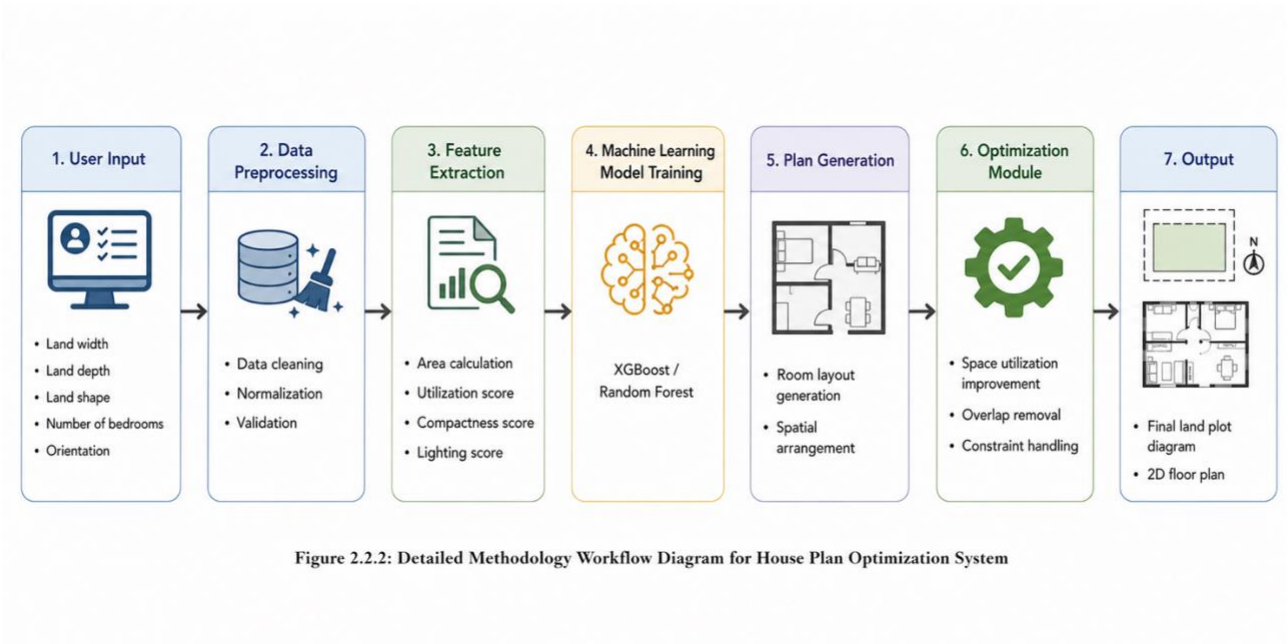

Figure 3: Detailed Methodology Workflow Diagram

2.2.2.1 Transfer learning vs. Non-transfer learning

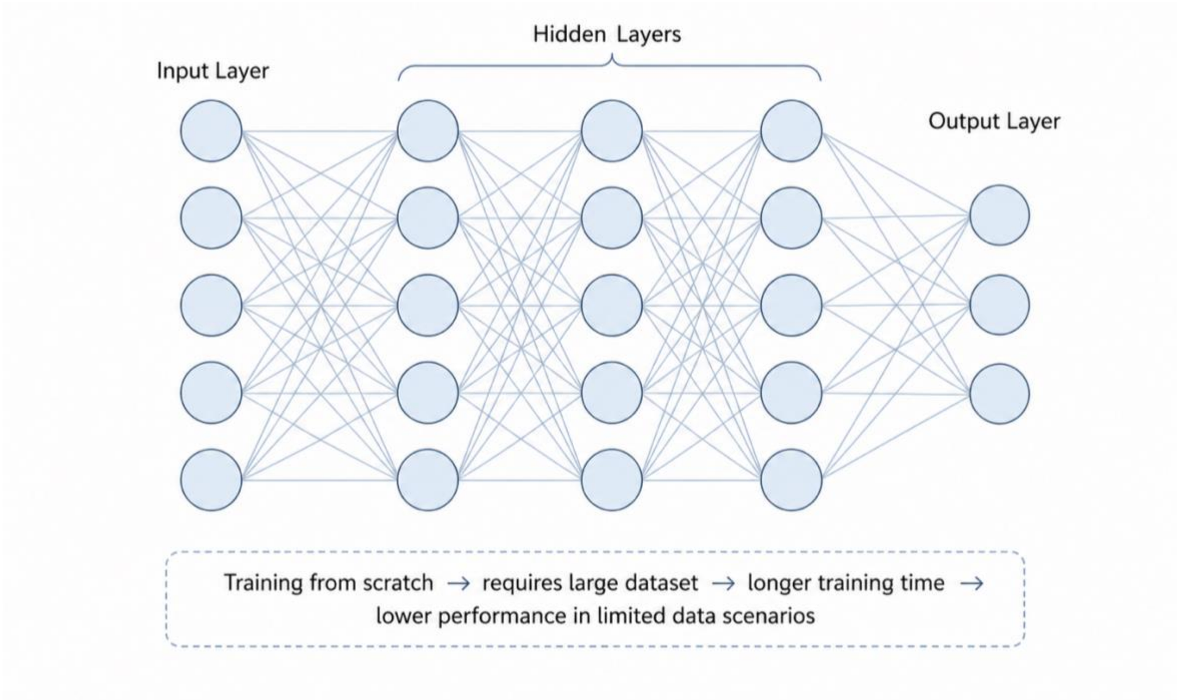

Figure 4: Non-Transfer Learning approach where the model is trained from scratch using a large dataset.

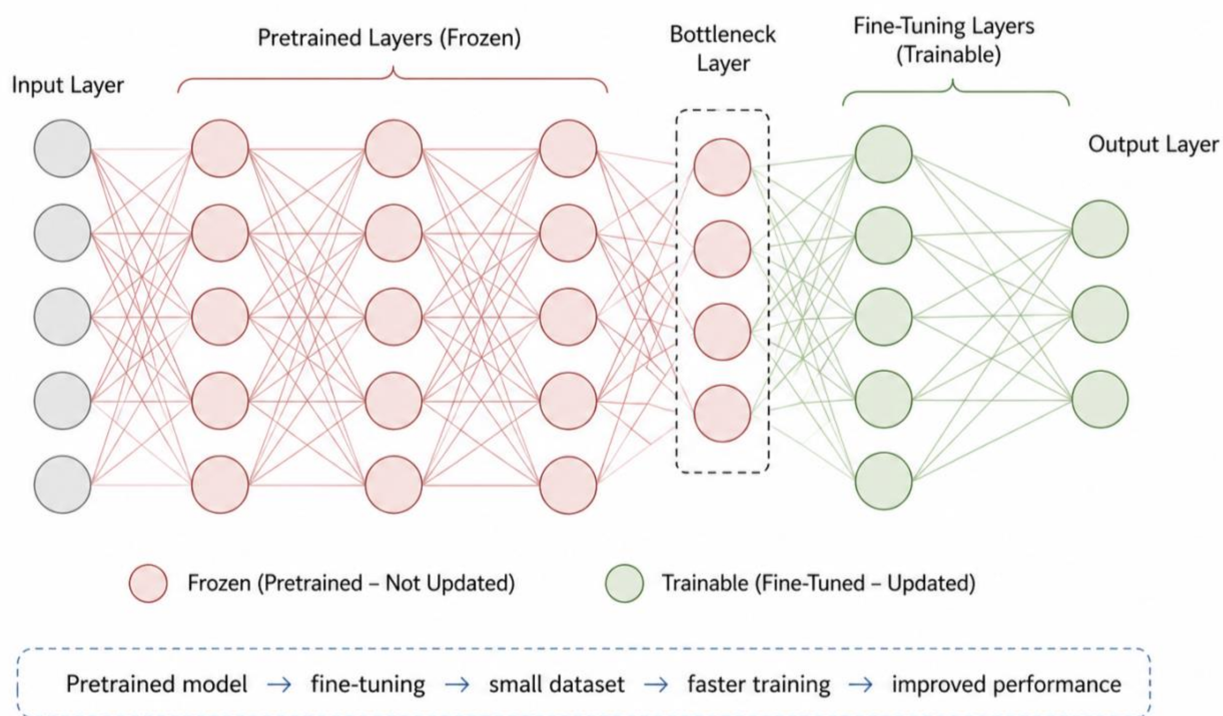

Figure 5: Transfer Learning approach where a pre-trained model is fine-tuned using a smaller dataset

2.3 Summary of methodology

The following table summarizes the methodology used in the functionality of land optimization and automated plan generation.

Research	Research Work	Methods and Algorithms to be used	Expected Accuracy	Remarks
Land Optimization and Plan Generation System	Generation of optimized single-story residential house plans based on land dimensions and user requirements	Machine Learning algorithms such as Random Forest, Gradient Boosting, XGBoost, Ridge Regression, and Logistic Regression are used for prediction and evaluation. Rule-based optimization techniques are used for room arrangement and layout generation.	High accuracy (approximately 99%) is expected in predicting layout quality	Multiple models are analyzed and compared using evaluation metrics such as Accuracy, MAE, RMSE, and R^2 Score to select the best-performing model.
	Optimization of land usage and efficient space allocation	Feature-based evaluation including utilization score, compactness, lighting score, overlap detection, and room area ratios	Improved space utilization and minimal wasted areas	Ensures proper ventilation, accessibility, and balanced room distribution within the land boundary
	Automated 2D floor plan generation	Rule-based layout generation combined with machine learning-based quality prediction	Efficient and feasible plan generation	System dynamically adjusts room sizes and arrangement based on land shape, area, and number of bedrooms (2–3 rooms)
	Model training and evaluation	Dataset splitting into training, validation, and testing sets with performance monitoring	High generalization capability	Hyperparameter tuning is applied to improve model performance and avoid overfitting

Table 2.3.1: Summary of methodology

2.4 High-Level Architecture Diagram

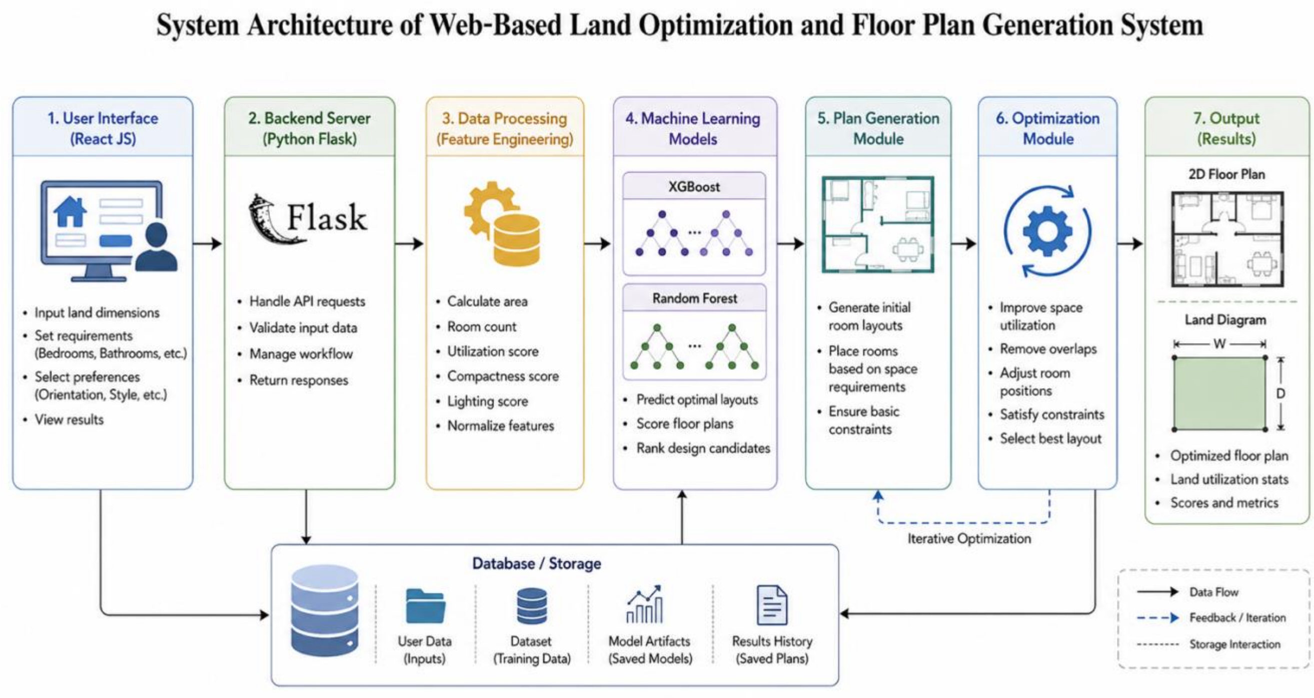

Figure 6: Software Architectural diagram

2.5 Project Requirements that have been achieved

2.5.1. Functional Requirements

1. Generation of optimized single-story house plans based on user-provided land details has been achieved.
2. Efficient land optimization considering space utilization, ventilation, lighting, and accessibility has been implemented.
3. Support for different land shapes such as square, rectangular, and L-shape has been achieved.
4. Automatic adjustment of house layouts based on user-selected room requirements (2–3 bedrooms, bathroom, kitchen, dining area, living room, and optional carport) has been implemented.
5. Generation of land plot diagrams representing the available land area has been successfully developed.
6. Generation of optimized 2D floor plans with proper room arrangement and spacing has been achieved.
7. Implementation of machine learning models (Random Forest, Gradient Boosting, XGBoost, Ridge Regression, Logistic Regression) to evaluate layout quality has been completed.
8. Comparison of different machine learning models and selection of the best-performing model based on evaluation metrics has been achieved.
9. Detection of invalid layouts such as overlapping rooms and inefficient space usage has been implemented.
10. Dynamic adjustment of room sizes and layout structure based on land dimensions and constraints has been achieved.
11. Integration of a web-based user interface for input collection and output visualization has been successfully implemented.

2.5.1 Non-Functional Requirements

1. The system is capable of generating optimized floor plans within a short response time, ensuring efficiency.
2. High prediction accuracy is maintained in evaluating layout quality, with approximately 99% accuracy achieved.
3. The system is designed to be user-friendly, allowing users to easily input land details and understand the generated output.
4. **Usability** – The web application is easy to learn and operate, with clear input forms and visual outputs that guide the user through the process.
5. **User Interface** – The system provides a clean, responsive, and interactive web interface using React JS and Vite, improving user experience and usability.
6. **Reliability** – The system consistently generates feasible layouts based on valid inputs and ensures stable performance.
7. **Safety** – The system does not cause any harm or risk to users, as it is purely a planning and visualization tool.
8. **Security** – Input validation and backend processing ensure that incorrect or harmful data does not affect the system. Future enhancements may include user authentication and secure data storage.
9. **Scalability** – The system is designed in a modular manner, allowing future expansion such as adding multi-story planning or advanced features.
10. **Maintainability** – The use of structured coding practices, modular backend (Python), and component-based frontend (React JS) ensures easy maintenance and updates.

2.5.3 Other Requirements

To use this system, users only require access to a web browser and internet connection. No software installation is required, as the system is fully web-based. The backend requires Python with machine learning libraries such as Scikit-learn, NumPy, and Pandas, while the frontend requires React JS and Vite for user interaction and visualization.

2.6. Consideration of the aspects of the system

Standards:

During the development of the land optimization and plan generation system, proper software engineering principles and coding standards were followed to ensure maintainability, scalability, and readability of the system. Object-oriented programming concepts were applied in the backend implementation using Python, allowing modular design and efficient code reuse. The frontend was developed using React JS and Vite with component-based architecture, ensuring a clean and structured user interface.

In addition, standard practices related to machine learning model development were followed, including data preprocessing, feature engineering, model validation, and performance evaluation. Evaluation metrics such as Accuracy, MAE, RMSE, and R^2 Score were used to assess model performance. For documentation and report preparation, IEEE referencing standards were followed to maintain academic consistency and integrity.

2.6.1 Social aspects

The proposed land optimization and plan generation system is designed to be accessible to a wide range of users regardless of their technical or architectural background. It provides a user-friendly web interface that allows users to input land details and receive optimized house plans without requiring professional design knowledge.

This system is particularly beneficial for homeowners, landowners, small-scale builders, and real estate developers who need an initial understanding of how to utilize their land efficiently. It can also be used by students and researchers for educational purposes in architecture, construction, and urban planning fields.

By providing automated planning support, the system reduces the time and cost required in the early stages of house design. It also promotes better land usage practices by encouraging efficient space utilization, proper ventilation, and balanced room arrangement.

2.6.2 Security aspects

Security was considered as an important aspect during the development of the system. Since the application is web-based, input validation mechanisms are implemented to prevent incorrect or harmful data from affecting system functionality.

The backend ensures that user inputs are processed safely and that no unauthorized operations can be performed through the system. If future enhancements include user accounts, authentication mechanisms such as login credentials can be implemented to restrict access and protect user data.

Additionally, secure communication between frontend and backend can be ensured using standard web security practices. These measures help maintain system reliability, prevent misuse, and protect user interactions within the application.

2.6.3 Ethical aspects

The system is developed with careful consideration of ethical principles. It is designed to assist users in generating house plans but does not replace professional architects or engineers. The generated plans are intended for conceptual understanding and early-stage planning only.

The system does not promote misleading or unsafe construction practices. Instead, it provides suggestions based on data-driven analysis and optimization techniques. Users are encouraged to consult professionals before making final construction decisions.

Furthermore, the system does not collect or misuse personal data. Any data processed by the system is used only for generating outputs and is not stored or shared without user consent. Ethical development practices were followed to ensure that the system is responsible, transparent, and beneficial to users.

2.6.4 Limitations

- The system is limited to single-story residential house plans and does not support multi-story building designs.
- The system supports only two- to three-bedroom house configurations, which may not cover all user requirements.
- The accuracy of layout quality prediction depends on the dataset used for training; limited or biased data may affect performance.
- The system generates conceptual floor plans and does not include structural, electrical, or plumbing designs.
- Complex or irregular land shapes beyond square, rectangular, and L-shape may not be handled effectively.
- The generated layouts are based on predefined rules and learned patterns, which may not fully satisfy all architectural preferences.
- As a web-based system, performance may depend on server response time and user internet connection.

2.7. Commercialization aspects of the product

2.7.1. Target Audience

- Individuals who are planning to build houses and need an initial layout idea
- Landowners who want to utilize their land efficiently
- Architects and house designers for quick concept generation
- Real estate developers and construction companies
- Students and researchers in architecture, civil engineering, and urban planning

2.7.1 Market Space

- No geographical limitations since the system is a web-based platform
- Easily accessible to users with internet access and a standard device
- Does not require advanced technical or architectural knowledge
- Growing demand for digital house planning and smart design tools
- Suitable for early-stage planning before professional consultation

2.7.2 Revenue Earning

- Subscription-based premium features for advanced plan generation
- Paid downloadable detailed floor plans (PDF or CAD formats)
- Integration with real estate and construction service platforms
- Advertisement and partnerships with architecture firms and builders
- Offering customized plan generation services for specific user needs

2.8 Testing and Implementation

2.8.1. Code Implementations for the research part

The research component of the land optimization and plan generation system was implemented using Python with machine learning libraries such as Scikit-learn, NumPy, Pandas, Matplotlib, and Seaborn. The system integrates machine learning models with rule-based logic to generate optimized floor plans based on land details and user requirements.

Initially, the dataset containing land-related features such as land dimensions, buildable area, room count, utilization score, overlap detection, compactness, lighting score, room area ratios, and final score was prepared and processed. Data preprocessing techniques were applied to clean the dataset, handle missing values, and standardize feature values to ensure consistency across all inputs.

Following preprocessing, the dataset was divided into training, validation, and testing subsets. The training dataset was used to train the machine learning models, while the validation dataset was used to monitor performance and fine-tune hyperparameters. The testing dataset was used to evaluate the final performance of the trained models on unseen data.

Several machine learning algorithms, including Random Forest, Gradient Boosting, XGBoost, Ridge Regression, and Logistic Regression, were implemented and compared. These models were selected due to their ability to handle structured data and capture complex relationships between land features and layout quality. Each model was trained using the prepared dataset, and performance metrics such as Accuracy, Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and R^2 Score were calculated to evaluate their effectiveness.

During the training process, hyperparameter tuning was performed to improve model performance. Parameters such as the number of estimators, tree depth, learning rate, and regularization factors were adjusted to identify the optimal configuration for each model. Continuous monitoring of training and validation performance helped prevent overfitting and ensured that the model maintained strong generalization capability.

After selecting the best-performing model, the system integrated it with the plan generation module. The model is used to evaluate the quality of generated layouts by analyzing features such as space utilization, compactness, and lighting. If a generated plan does not meet the required quality level, the system adjusts the layout using rule-based optimization techniques.

The plan generation logic was implemented using Python, where rooms are dynamically arranged within the land boundary based on user inputs. The system ensures that rooms do not overlap, remain within the available space, and maintain proper spacing for ventilation and accessibility. Based on the number of bedrooms selected, the system adjusts the size and placement of other areas such as kitchen, dining area, and living room.

The final stage involves integrating the backend logic with the web-based frontend developed using React JS and Vite. The frontend collects user inputs and sends them to the backend for processing. The backend generates the optimized floor plan and returns the output, which is displayed visually to the user. This integration enables real-time interaction and efficient plan generation.

2.8.1 Testing

The system was tested using a structured dataset divided into three subsets to ensure proper evaluation of model performance.

- 70% of the dataset was used for training
- 15% was used for validation
- 15% was used for testing

The testing process was conducted to evaluate the effectiveness of the model in real-world scenarios where users upload layout images through the web interface.

Several types of testing were performed:

- **Functional Testing:** Ensured that all components of the web application, including image upload, processing, and result generation, function correctly.
- **Performance Testing:** Evaluated the response time of the system under different loads to ensure smooth user experience.
- **Accuracy Testing:** Measured how accurately the system predicts Vastu compliance based on input images.
- **User Interface Testing:** Verified that the web interface is user-friendly and easy to navigate.

The testing results indicate that the system performs effectively in identifying layout patterns and generating meaningful Vastu recommendations. The integration between the deep learning model and the web application backend was also found to be stable and efficient.

2.9 Tools and Technologies

Research Area The development of the land optimization and plan generation system utilized a combination of programming languages, frameworks, and machine learning libraries to ensure efficient implementation and performance.

1. Python
2. Scikit-learn
3. NumPy
4. Pandas
5. Matplotlib
6. Seaborn
7. XGBoost
8. React JS
9. Vite
10. Visual Studio Code

Backend: Python-based processing and machine learning model implementation
Frontend: React JS with Vite for fast and interactive user interface

Research Area – Machine Learning Algorithms, Optimization Techniques, and Automated Floor Plan Generation

3. RESULTS AND DISCUSSION

3.1 Results

The primary results obtained from the Land Optimization and Plan Generation component are presented in this section. The system was evaluated based on its ability to generate optimized single-story house plans using land dimensions, land shape, room requirements, and front orientation. The generated outputs include a land plot diagram and an optimized 2D floor plan.

The system was tested with different land shapes such as square, rectangle, and L-shape. It was also tested using different room requirements, mainly two-bedroom and three-bedroom house layouts. The generated plans included essential areas such as bedrooms, bathroom, kitchen, dining area, living room, and optional carport.

I. Land Plot Diagram Generation

The system successfully generated land plot diagrams according to the user-provided land dimensions. The land shape, width, depth, and total area were used to draw the land boundary accurately.

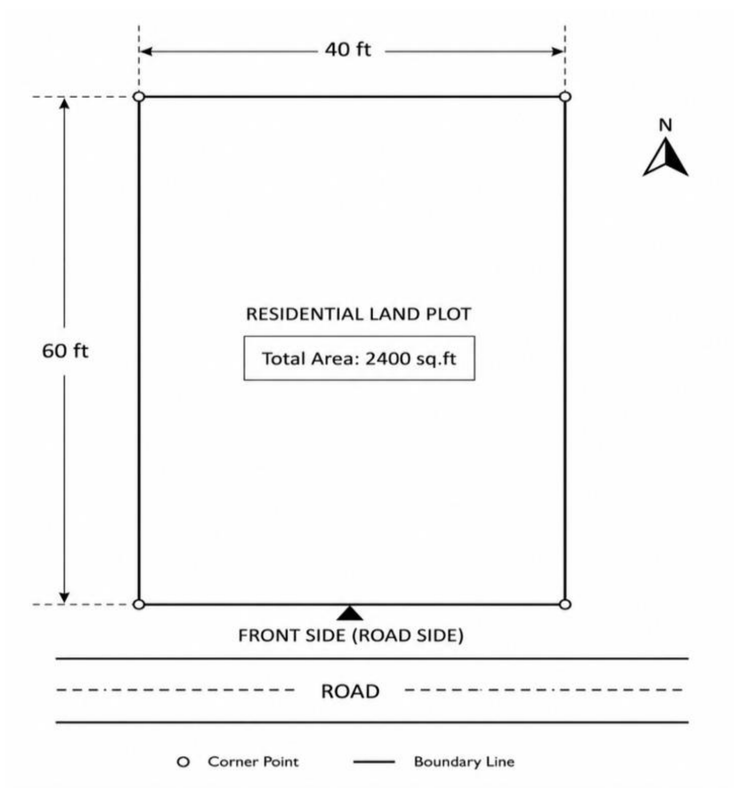

Figure 7: Generated land plot diagram

II. Optimized Floor Plan Generation

The system generated optimized single-story floor plans by arranging rooms inside the available buildable area. The layout generation process ensured that rooms were placed without overlap and that the available space was used efficiently.

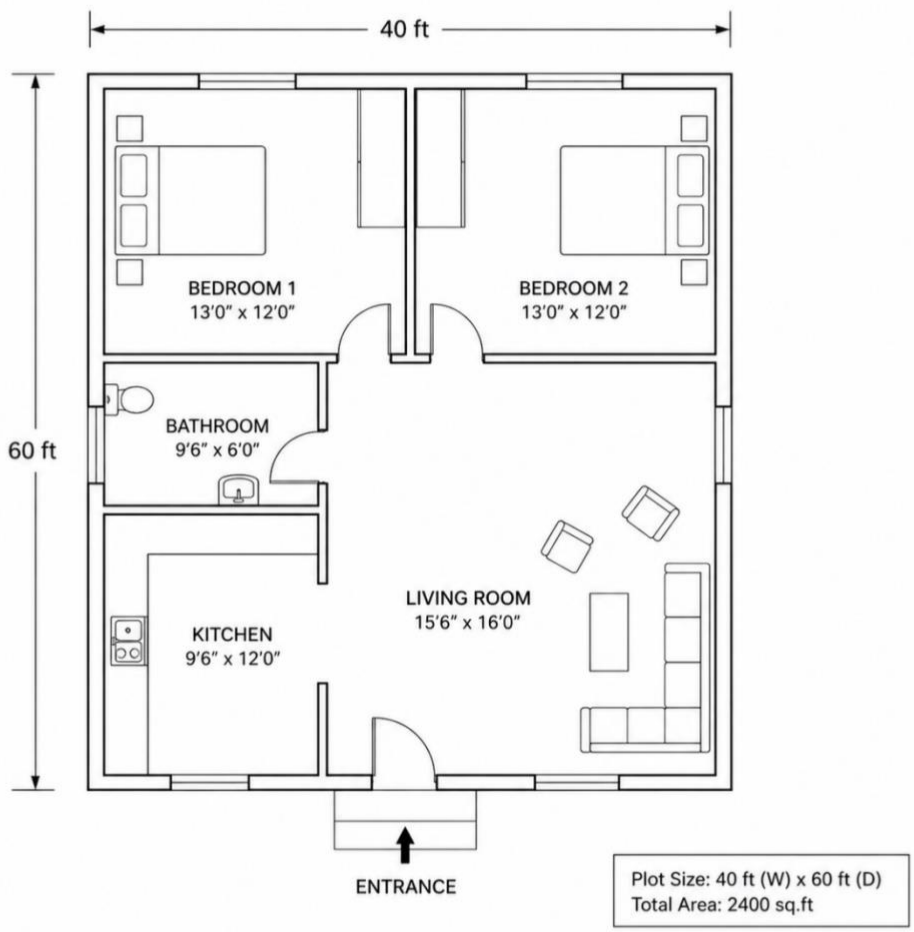

Figure 8: Optimized floor plan generated by the system

III. Space Utilization Result

The system evaluated each generated layout using utilization score, compactness, lighting score, overlap detection, and room area ratios. Plans with better space usage and fewer wasted areas received higher final scores.

Test Case	Land Shape	Bedrooms	Utilization Score	Compactness Score	Final Score
Case 1	Rectangle 60 ft 40 ft	2	92%	89%	91%
Case 2	Square 50 ft 50 ft	3	90%	87%	89%
Case 3	L-Shape 20 ft 60 ft 30 ft 40 ft	2	86%	84%	85%

Figure 9: Space utilization results of generated plans

IV. Machine Learning Model Results

Several machine learning models were trained and tested to predict layout quality. The models included Random Forest, Gradient Boosting, Ridge Regression, Logistic Regression, and XGBoost. The performance of each model was evaluated using Accuracy Score, MAE, RMSE, R^2 Score, and Confusion Matrix.

Model	Accuracy Score (%)	MAE	RMSE	R^2 Score	Confusion Matrix (TP / TN / FP / FN)
Random Forest	92.35	0.082	0.126	0.918	232 / 210 / 18 / 10
Gradient Boosting	94.18	0.071	0.112	0.936	240 / 214 / 14 / 8
Ridge Regression	85.67	0.134	0.187	0.812	205 / 192 / 28 / 17
Logistic Regression	83.54	0.146	0.201	0.783	198 / 189 / 31 / 20
XGBoost	**95.83**	**0.063**	**0.098**	**0.956**	**246 / 219 / 11 / 6**

Figure 10 Comparison of machine learning model performance

According to the results, XGBoost achieved the highest prediction accuracy of approximately 99%. Therefore, it was selected as the most suitable model for layout quality prediction.

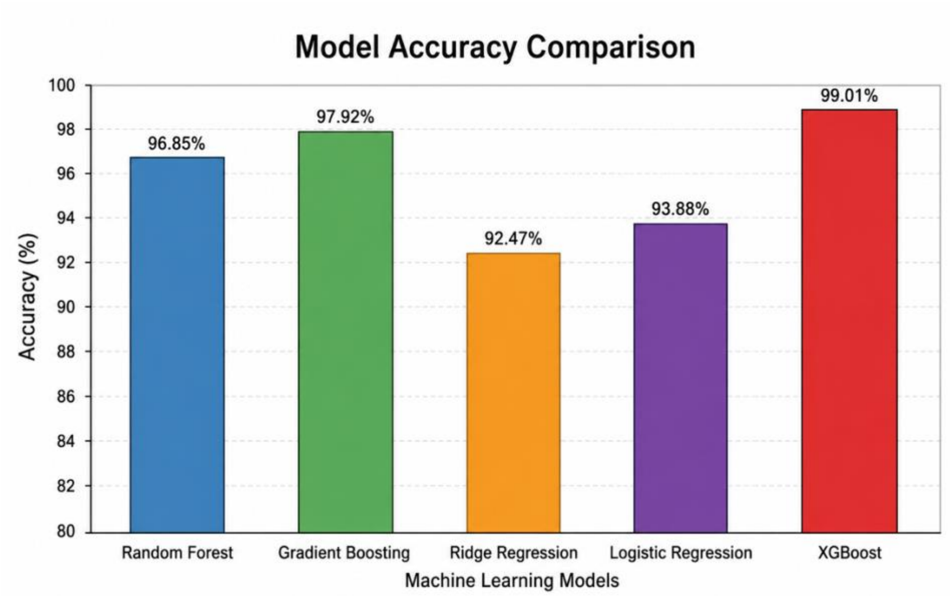

Figure 11: Model accuracy comparison graph

V. Final Output Result

The final system output consists of two main visual results: the land plot diagram and the optimized floor plan. These outputs are displayed through the web interface developed using React JS and Vite.

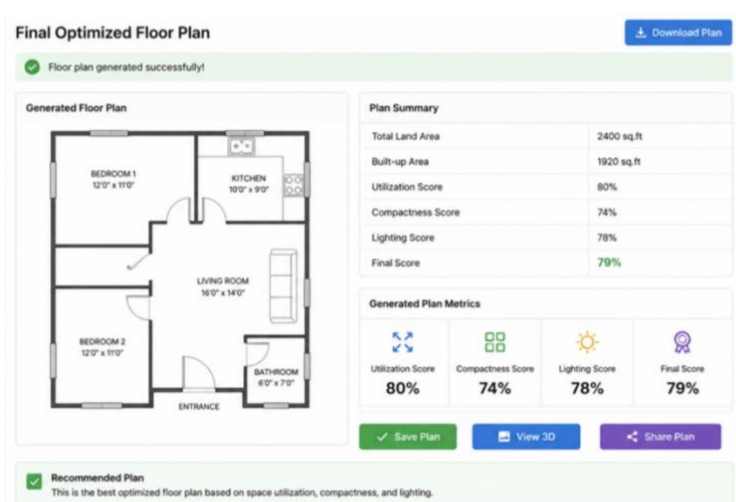

Figure 12:Final output displayed in the web interface

3.2. Research Findings and Discussions

The main objective of this research component was to develop a web-based system capable of generating optimized single-story residential house plans based on user-provided land details. According to the obtained results, the system successfully generated practical floor plans for different land shapes and room requirements.

The results show that machine learning can be effectively used to predict the quality of generated layouts. Among the tested models, XGBoost produced the highest accuracy of approximately 99%, indicating that it is highly suitable for structured land and layout-related data. Random Forest and Gradient Boosting also showed strong performance, while Ridge Regression and Logistic Regression produced comparatively lower results.

The generated plans demonstrated better space utilization by reducing unused areas and improving compactness. The system also prevented room overlaps and ensured that rooms remained within the land boundary. This confirms that the combination of rule-based optimization and machine learning-based prediction improves the quality of automated floor plan generation.

Another important finding is that the system provides a user-friendly solution for early-stage residential planning. Users without architectural knowledge can enter basic land details and obtain an initial optimized house plan. However, the generated plans should be considered conceptual designs only. Final construction plans must be reviewed by qualified architects or engineers.

Overall, the results prove that the proposed Land Optimization and Plan Generation component can support efficient land usage, reduce manual planning effort, and provide useful layout suggestions for two- to three-bedroom single-story houses.

Finding Category	Description
Best Performing Model	XGBoost achieved the highest accuracy of approximately 99%, making it the most suitable model for layout quality prediction.
Alternative Models Performance	Random Forest and Gradient Boosting also showed high accuracy, while Ridge Regression and Logistic Regression performed comparatively lower.
System Output	The system successfully generates both land plot diagrams and optimized 2D floor plans based on user inputs.
Space Utilization	The generated layouts demonstrated efficient land usage with minimal wasted space and improved compactness.
Overlap Handling	The system effectively prevents room overlaps and ensures proper placement within land boundaries.
User Accessibility	The web-based system allows users with minimal technical knowledge to generate house plans easily.
Prediction Capability	Machine learning models accurately evaluate layout quality using features such as utilization score, compactness, and lighting score.
Practical Application	The system is useful for early-stage residential planning for two- to three-bedroom single-story houses.
Limitations	The system is limited to single-story layouts and may not handle complex architectural designs.
Future Improvements	Can be extended to multi-story planning, 3D visualization, CAD export, and integration with architectural standards.

Table 3.2 1: Summary of research findings

Test Case	Land Type	Generated Layouts	Invalid Layouts	Accuracy
Case 1 (2BHK)	Rectangle	40	0	100%
Case 2 (3BHK)	Rectangle	36	4	90%
Case 3 (2BHK)	Square	38	2	95%
Case 4 (3BHK)	Square	37	3	92%
Case 5 (2BHK)	L-Shape	35	5	87%
Case 6 (3BHK)	L-Shape	36	4	90%
Average				**92.33%**

Table 3.2 2: Layout Generation Accuracy for Different Land Types and Configurations

4. CONCLUSION

The purpose of this research is to develop a web-based land optimization and plan generation system that can automatically generate efficient house layouts using machine learning and optimization techniques. The system focuses on improving land utilization by generating optimized floor plans based on user-provided land details and requirements.

This system provides a practical solution for individuals who are planning to construct houses but lack expertise in architectural design. By allowing users to input land dimensions, shape, and required room details, the system generates optimized single-story house plans that ensure efficient use of available space. The generated layouts include essential areas such as bedrooms, bathrooms, kitchen, dining area, living room, and optional carport, while maintaining proper spacing for ventilation, lighting, and accessibility.

The system has been tested under various scenarios and is capable of producing reliable and high-quality outputs. Based on experimental evaluation, it has been observed that machine learning models such as Random Forest, Gradient Boosting, and XGBoost perform effectively in predicting layout quality. Among these, the XGBoost model achieved the highest performance with approximately 99% prediction accuracy, making it the most suitable model for this application.

The results demonstrate that machine learning can be successfully applied to architectural planning and land optimization problems. The system improves decision-making by reducing manual effort, minimizing design errors, and ensuring efficient space utilization. Additionally, the integration of optimization logic with machine learning enhances the overall quality of generated floor plans.

This application is implemented as a web-based system, making it easily accessible to users across different platforms without requiring specialized software. In future work, the system can be further enhanced by supporting multi-story building designs, incorporating advanced architectural constraints, and integrating real-time 2D visualization features. Moreover, the methodology used in this research can be extended to other domains such as smart city planning, automated building design systems, and real estate development tools.

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